

OOPS – ASSIGNMENT – 04

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1.

```
#include <iostream>
```

```
using namespace std;
```

```
class Rectangle
```

```
{
```

```
public:
```

```
    int length, breadth;
```

```
    Rectangle()
```

```
{
```

```
    length = 0;
```

```
    breadth = 0;
```

```
}
```

```
    Rectangle(int l, int b)
```

```
{
```

```
    length = l;
```

```
    breadth = b;
```

```
}
```

```
    Rectangle(int x)
```

```
{
```

```

        length = x;
        breadth = x;
    }

    int area()
    {
        return length * breadth;
    }
};

```

```

int main()
{
    Rectangle r1;
    Rectangle r2(5);
    Rectangle r3(4, 6);

    cout << r1.area() << endl;
    cout << r2.area() << endl;
    cout << r3.area() << endl;

    return 0;
}

```

```

cd "/Users/vihaan/Desktop/DSA/10.GRAPHS/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPHS/"trash
● (base) vihaan@Vihaans-Macbook 10.GRAPHS % cd "/Users/vihaan/Desktop/DSA/10.GRAPHS/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPHS/"trash
0
25
24
○ (base) vihaan@Vihaans-Macbook 10.GRAPHS %

```

2.

```

#include <iostream>

using namespace std;

```

```
class Rectangle
{
public:
    int length, breadth;

    Rectangle()
    {
        length = 0;
        breadth = 0;
    }

    Rectangle(int l, int b)
    {
        length = l;
        breadth = b;
    }

    Rectangle(int x)
    {
        length = x;
        breadth = x;
    }

    int area()
    {
        return length * breadth;
    }
}
```

```

    }

    ~Rectangle()
    {
    }

};

int main()
{
    Rectangle r[3] = {Rectangle(), Rectangle(5), Rectangle(4, 6)};

    for (int i = 0; i < 3; i++)
    {
        cout << r[i].area() << endl;
    }

    return 0;
}

```



```

(base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash
0
25
24
(base) vihaan@Vihaans-Macbook 10.GRAPH5 %

```

3.

```

#include <iostream>

using namespace std;

```

```

class Test
{

```

```

public:
    Test()
    {
        cout << "Constructor called" << endl;
    }

    ~Test()
    {
        cout << "Destructor called" << endl;
    }
};

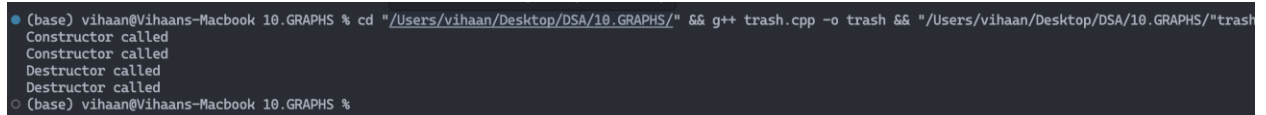
```

```

int main()
{
    Test t1;
    Test t2;

    return 0;
}

```



```

• (base) vihaan@Vihaans-Macbook 10_GRAPH % cd "/Users/vihaan/Desktop/DSA/10_GRAPH/10_GRAPHS/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10_GRAPH/10_GRAPHS/'trash
Constructor called
Constructor called
Destructor called
Destructor called
• (base) vihaan@Vihaans-Macbook 10_GRAPH %

```

4.

```
#include <iostream>
```

```
using namespace std;
```

```
class Test
```

```
{  
public:  
    int x;  
    Test()  
    {  
        x = 10;  
    }  
};
```

```
int main()  
{  
    int *a = new int;  
    *a = 5;  
    cout << *a << endl;  
    delete a;
```

```
    float *b = new float;  
    *b = 3.5;  
    cout << *b << endl;  
    delete b;
```

```
    int *arr1 = new int[3];  
    arr1[0] = 1;  
    arr1[1] = 2;  
    arr1[2] = 3;  
    for (int i = 0; i < 3; i++)  
        cout << arr1[i] << " ";
```

```
cout << endl;
```

```
delete[] arr1;
```

```
float *arr2 = new float[2];
```

```
arr2[0] = 1.1;
```

```
arr2[1] = 2.2;
```

```
for (int i = 0; i < 2; i++)
```

```
    cout << arr2[i] << " ";
```

```
cout << endl;
```

```
delete[] arr2;
```

```
Test *t = new Test;
```

```
cout << t->x << endl;
```

```
delete t;
```

```
Test *tarr = new Test[2];
```

```
for (int i = 0; i < 2; i++)
```

```
    cout << tarr[i].x << " ";
```

```
cout << endl;
```

```
delete[] tarr;
```

```
return 0;
```

```
}
```

```
● (base) vihaan@Vihaans-Macbook 10.GRAPH5 % cd "/Users/vihaan/Desktop/DSA/10.GRAPH5/" && g++ trash.cpp -o trash && "/Users/vihaan/Desktop/DSA/10.GRAPH5/"trash
5
3.5
1 2 3
1.1 2.2
10
10 10
○ (base) vihaan@Vihaans-Macbook 10.GRAPH5 %
```